# ECM–TQAG: Evidence-Contracted Multimodal Generation of Textbook Question–Answer Items from Scanned Documents

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***Abstract*—We present ECM–TQAG, a method for generating evidence-grounded multimodal question–answer items from scanned Vietnamese-law documents. The method binds every item to a per-chunk conditioning bundle of recognised text, document structure, and a hash-identified image census; an evidence contract then requires the generator to select its text quotation and source image, to plan the answer before wording the question, and to keep that evidence tuple fixed while realising the item. A deterministic verifier admits a candidate only if the returned tuple is reproducible against the bundle, and two arm-blind judges supply the semantic criteria of a derived validity endpoint. We evaluate the contract in a paired three-arm census of image-bearing chunks in which the prompt is the only quantity that varies: the generator, response schema, decoding settings, and question-type assignment are held fixed, so a paired contrast isolates the contract. All failures remain in the intention-to-test denominator, and the two confirmatory contrasts use exact McNemar tests under Holm control. The census shows that the contract improves structural compliance while end-to-end valid yield stays low for every arm, with the visual necessity of source images as the dominant bottleneck.**

***Index Terms*—multimodal question generation, document intelligence, evidence grounding, structured decoding, legal education**

## I. Introduction and Related Work

Scanned textbooks combine prose, diagrams, hierarchies, and tables, motivating textbook QA [1] and its generation counterpart. Yet a fluent question may assert unsupported content [2] or may merely appear multimodal while remaining answerable from text alone, a phenomenon analogous to language shortcuts in VQA [3], [4]. Existing document models combine text, layout, and pixels [5]–[7] or read pixels directly [8], [9], while DocVQA and ChartQA assess answers [10], [11]. Educational and visual question generation, in turn, motivate our construction task [12]–[14].

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Our contribution is the ECM–TQAG method itself: evidence support and visual dependence are stated as a contract on the generation call, and that contract is made checkable by a deterministic verifier that recomputes the declared evidence against the source. The method is therefore specified together with the endpoint it is measured by, and it is designed so that a paired experiment can attribute any difference in that endpoint to the contract alone. Section II formalises the method, and Section III instantiates it as a bounded census on Vietnamese-law material.

## II. Method

ECM–TQAG comprises a conditioning bundle that fixes what may serve as evidence, an evidence contract that constrains how an item is produced from that bundle, and a deterministic verifier that turns each returned item into a binary outcome. Fig. 1 shows the composition.

### A. Conditioning bundle

Each source chunk $c$ is represented by

$$E_c = \big(T_c,\ L_c,\ \mathcal{I}_c\big), \tag{1}$$

where $T_c$ is the recognised text, $L_c$ the document structure, and $\mathcal{I}_c$ a census of page images identified by content hash. Nothing outside $E_c$ is admissible evidence for a chunk, which is what makes the contract checkable rather than declarative.

### B. Contracted generation

A single multimodal generator $g$ is fixed for the whole method. For arm $a$ in the condition set $\mathcal{A} = \{\text{ECM}, \text{DIR}, \text{STR}\}$, one call produces one candidate,

$$y_{c,a} = g\big(\pi_a,\ E_c\big), \tag{2}$$

where $\pi_a$ is the arm prompt. Source bytes, decoding settings, question type $t_c$, and the type-conditioned response schema are identical across arms, so $\pi_a$ is the only quantity in (2) that varies; the direct arm requests one multimodal item and the structured no-contract arm additionally emphasises exact schema completion.

The ECM contract $\pi_{\text{ECM}}$ adds four operations to that shared instruction. It selects an exact quotation $q$ from $T_c$ together with an image $h$ from $\mathcal{I}_c$; it plans a unique answer $\alpha$ before the question is worded; it holds the evidence tuple

$$\tau = (t,\ q,\ \alpha,\ h) \tag{3}$$

fixed while the item is realised; and it self-checks multimodal necessity and schema conformance. These are instructions inside one call, not separately observed stages. The provider returns only the final structured item, so hidden intermediate reasoning is never treated as observed evidence, and we write $\tau(y)$ for the tuple carried by a response $y$.

### C. Deterministic verification

Verification recomputes $\tau(y)$ against $E_c$. The gate is the conjunction

$$\begin{aligned} G(y, E_c) = \mathbb{1}\{t = t_c\} \cdot \mathbb{1}\{q \sqsubseteq T_c\} \cdot \mathbb{1}\{h \in \mathcal{I}_c\} \\ \cdot\, \mathbb{1}\{\sigma(y)\} \cdot \mathbb{1}\{\mu(y)\}, \end{aligned} \tag{4}$$

where $q \sqsubseteq T_c$ requires $q$ to be a contiguous substring of the recognised text, $\sigma$ is exact conformance to the response schema with a nonempty question, answer, visual description, and necessity rationale, and $\mu$ requires, for multiple choice, four distinct options with a valid unique answer index whose option equals $\alpha$. The code accepts exactly one JSON object and performs no insertion, aliasing, translation, normalisation, coercion, or repair, so $G$ is a property of the response rather than of any post-processing. Passing $G$ establishes mechanical source consistency only, not truth, legal correctness, pedagogical value, or genuine visual necessity.

### D. Derived endpoint

Two arm-blind judges $j \in \{1, 2\}$ from independent model families supply the semantic criteria. Each returns the same seven-field object: five integer scores $s_{jk} \in \{1, \ldots, 5\}$, a critical-provenance flag $p_j$, and a nonempty rationale. With $k \leq 3$ indexing evidence correctness, visual necessity, and answerability, the primary endpoint is

$$\begin{aligned} V(c, a) = G(y_{c,a}, E_c) \cdot \mathbb{1}\Big\{\min_{j,k \leq 3} s_{jk} \geq 3\Big\} \\ \cdot \prod_{j=1}^{2} \mathbb{1}\{\gamma_j\}\mathbb{1}\{p_j = 0\}, \end{aligned} \tag{5}$$

where $\gamma_j$ is the judge record passing its own identity, schema, and local gates. Every factor in (5) is computed by code; caller-supplied validity flags are never analysis inputs. A missing, transport-invalid, identity-invalid, or schema-invalid record leaves the corresponding factor at zero, so $V(c, a) = 0$ under intention-to-test rather than being removed from the denominator.

TABLE I
GATE ADMISSION PER ARM, 16 CHUNKS EACH.

| Arm | Admitted | Rate |
|---|---|---|
| ECM full | 6/16 | 0.375 |
| Direct | 5/16 | 0.313 |
| Structured no-contract | 3/16 | 0.188 |

### E. Identification

Because $g$, $E_c$, $t_c$, the schema, and the decoding settings are common to all arms, the paired difference $V(c, \text{ECM}) - V(c, a')$ for $a' \neq \text{ECM}$ varies only with the prompt. Each chunk therefore acts as its own control, and the contract is tested by exact paired comparisons over chunks with prespecified multiplicity control.

## III. Experimental Setup

The analysis frame is a bounded census of 16 Vietnamese-law image-bearing chunks, not a population sample. The generator is Qwen3-VL-8B-Instruct, held fixed across all three arms so that no model change can confound the prompt contrast, and the two arm-blind judges are Claude Sonnet 5 and GPT-5.6 Terra. Multiple-choice distractor integrity is enforced by the construction gate $\mu$ in (4) rather than by a judge field.

The primary endpoint is the intention-to-test valid yield $V$ of (5): every assigned chunk remains in the denominator, and schema, transport, missing-response, and gate failures count as invalid while being tabulated separately. The two confirmatory contrasts compare ECM against each control arm with exact McNemar tests under Holm control at $\alpha = 0.05$, with Clopper–Pearson intervals for per-arm yields. Five secondary judged vectors, namely evidence correctness, visual necessity, answerability and uniqueness, pedagogical value, and Vietnamese language quality, are reported separately rather than averaged, and they remain exploratory until calibrated against blinded human ratings.

## IV. Results

All 48 generation attempts of the $16 \times 3$ frame were executed, and the 14 candidates admitted by the gate $G$ were referred to both judges, giving 28 judging attempts of which 27 returned a schema-valid record; the remaining attempt was a terminal failure and is retained as such.

### A. Gate admission

Table I reports per-arm admission under (4). The ECM contract admitted the most candidates and the structured no-contract prompt the fewest.

### B. Primary endpoint

Under the derived endpoint (5), a single candidate across all arms was valid, in the direct arm on a short-answer chunk whose source text was self-contained. Table II gives the intention-to-test yields with Clopper–Pearson intervals.

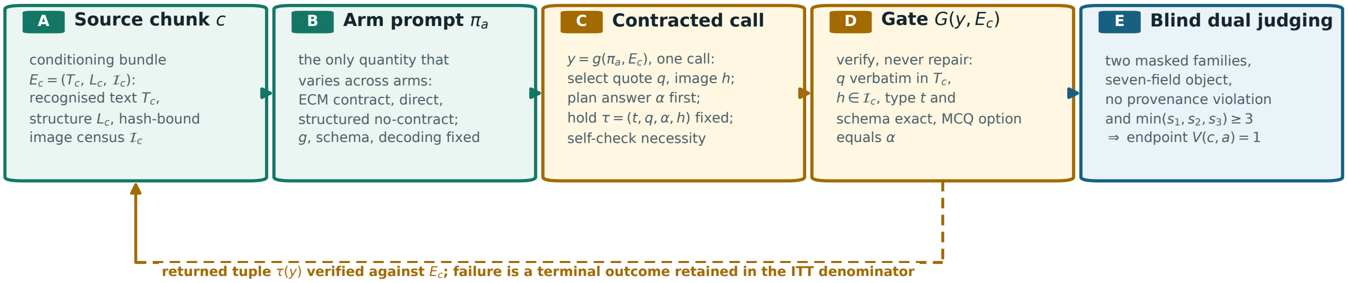


Fig. 1. ECM–TQAG. A per-chunk conditioning bundle $E_c$ conditions one generation call per arm; the arm prompt $\pi_a$ is the only quantity that varies. The gate $G$ recomputes the returned tuple $\tau(y)$ against $E_c$ without repair, and blind dual judging supplies the semantic factors of the derived endpoint $V(c, a)$.

TABLE II
INTENTION-TO-TEST VALID YIELD PER ARM, $n = 16$.

| Arm | Valid | Rate | 95% CP CI |
|---|---|---|---|
| ECM full | 0/16 | 0.000 | [0.000, 0.206] |
| Direct | 1/16 | 0.063 | [0.002, 0.302] |
| Structured no-contract | 0/16 | 0.000 | [0.000, 0.206] |

TABLE III
EXACT MCNEMAR CONTRASTS, HOLM-ADJUSTED AT $\alpha = 0.05$.

| Contrast | Discordant | $p$ | Holm | Reject |
|---|---|---|---|---|
| ECM vs. Direct | 1 | 1.000 | 0.025 | No |
| ECM vs. Structured | 0 | 1.000 | 0.050 | No |

### C. Confirmatory contrasts

Neither exact McNemar contrast rejected its null under Holm control (Table III). The one discordant pair favoured the direct arm; the ECM versus structured contrast had no discordant pair, so $p = 1.000$ by definition.

### D. Failure structure

Of the 34 terminal generation failures, 32 violated the output contract, 1 returned unparseable content, and 1 returned no response. The dominant violation was a quotation drawn from text visible in the page image but absent from $T_c$, which fails $q \sqsubseteq T_c$; the second most frequent was a multiple-choice answer reported as an option letter rather than the option text required by $\mu$.

### E. Judged vectors

Table IV reports mean judged scores over admitted candidates. These are conditional on admission and exploratory.

Evidence correctness, answerability, and language quality were high in every arm, whereas visual necessity was low throughout, with arm means between 1.80 and 2.67. Judges flagged a critical provenance violation in two records.

### F. Instrument robustness

A prospective second round re-executed the same frame after two changes made for measurement validity, neither of which alters $G$ or $V$: the judge families were substituted,

TABLE IV
MEAN JUDGED SCORES BY ARM, ADMITTED CANDIDATES ONLY.

| Dimension | ECM | Direct | Struct. |
|---|---|---|---|
| Evidence correctness | 4.18 | 3.80 | 4.33 |
| Visual necessity | 2.45 | 1.80 | 2.67 |
| Answerability | 4.64 | 4.80 | 4.67 |
| Pedagogical value | 3.09 | 3.20 | 3.33 |
| Vietnamese language | 4.73 | 4.50 | 4.83 |

because a provider outage had voided an intervening judging stage, and the prompt was corrected to state the option-indexing convention that the frozen contract left unspecified. Re-deriving the retained responses showed that 14 of 24 multiple-choice attempts had used one-based indices against a verifier requiring zero-based ones, so part of the round-one failure mass was an artefact of our specification rather than a generator limit.

Under the corrected specification, admission rose in every arm, to 12/16 (ECM), 7/16 (direct), and 10/16 (structured no-contract), and the ECM margin over the structured prompt narrowed from 6-vs-3 to 12-vs-10. The endpoint stayed low at 1/16, 0/16, and 1/16 respectively, with one and two discordant pairs and $p = 1.000$ for both contrasts. Over 58 judged assessments, visual necessity remained the binding constraint, with arm means of 2.00, 1.21, and 1.50 and no critical provenance violation.

## V. DISCUSSION

No arm reached a high end-to-end yield: the best observed rate was 1/16 in the direct arm, and neither confirmatory contrast rejected its null. The intervention therefore separates cleanly into two effects with opposite signs.

*a) The contract acts on structure.:* Admission under $G$ was highest for ECM at 6/16 against 5/16 and 3/16, so fixing the tuple $\tau$ before realisation does improve mechanical conformance to the evidence contract. This is the effect the contract is designed to produce, and it is the effect it produces.

*b) The contract does not act on visual dependence.:* The advantage did not survive to the endpoint, because $V$ additionally requires $\min(s_1, s_2, s_3) \geq 3$ and judged visual necessity was low in every arm, with means between 1.80

and 2.67. A prompt can require the generator to declare that an image is necessary; it cannot make the image necessary. This bottleneck is a property of the corpus, not of $\pi_a$: across four complete Vietnamese-law textbooks classified page by page, only 4 of 2167 pages carry a diagram at all (0.185%, 95% exact interval $[0.050\%, 0.472\%]$), and screening vector-rich candidates in the wider collection raises the yield only to 31 of 437 (7.1%, $[4.9\%, 9.9\%]$). Diagram presence is necessary but not sufficient for visual necessity, so these rates bound from above the fraction of this material on which the criterion could ever be met. Admitting any image-bearing chunk is therefore too weak a selection rule: it must be strengthened to chunks whose images carry information absent from $T_c$.

*c) Failures are generator-side, not design-side.:* Both dominant violations, quoting image-rendered text absent from $T_c$ and returning an option letter instead of the option text, are capability limits of Qwen3-VL-8B under a strict schema with no repair. The gate detects them as specified, which is why they appear as terminal outcomes rather than as silently corrected items.

*d) How much of the structural effect was measurement.:* The replication of §IV-F revises the reading of the structural result. Once the indexing convention was stated, admission improved in every arm and the ECM margin over the structured prompt fell from 6-vs-3 to 12-vs-10, so a substantial part of the round-one structural advantage reflected an unstated convention in our own contract rather than the intervention. The endpoint conclusion instead replicated: under the corrected specification and two different judge families, valid yield stayed at or below 1/16 in every arm and judged visual necessity stayed low. A first attempt at the same repair had to be discarded, because a pre-declared validity check found that a long clarification appended to all three arms induced arm-dependent decoder degeneration and inflated the ECM margin; strict verifiers are therefore sensitive to prompt mass as well as to prompt content.

*e) Interpretation boundaries.:* The census is bounded at 16 chunks and supports no population claim. Non-significance is not equivalence: at $\alpha = 0.05$ with $n = 16$, six unidirectional discordant pairs are required for significance, and the 0 to 2 pairs observed across both rounds provide evidence in neither direction. The judged vectors are exploratory until calibrated against blinded human ratings, and no deterministic check substitutes for expert legal or pedagogical review. The two actionable targets are the verbatim-quotation bottleneck and corpus selection by genuine visual dependence.

## VI. Conclusion

ECM–TQAG states evidence support and visual dependence as a contract on one generation call and makes that contract checkable by recomputing the declared evidence tuple against the source bundle, with a derived intention-to-test endpoint. In a paired three-arm census that varies only the prompt, the contract raised structural compliance but produced no significant advantage in complete-pipeline yield, and the binding constraint was the visual necessity of the source images rather than the contract itself. The method is thus executable and structurally beneficial, while end-to-end valid multimodal yield on this corpus and generator remains an open problem.

## Data Availability

The evidence consists of the frozen protocol, the evaluation instrument, and the complete census records with their primary analysis. Copyrighted page images and raw provider responses are not distributed.


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